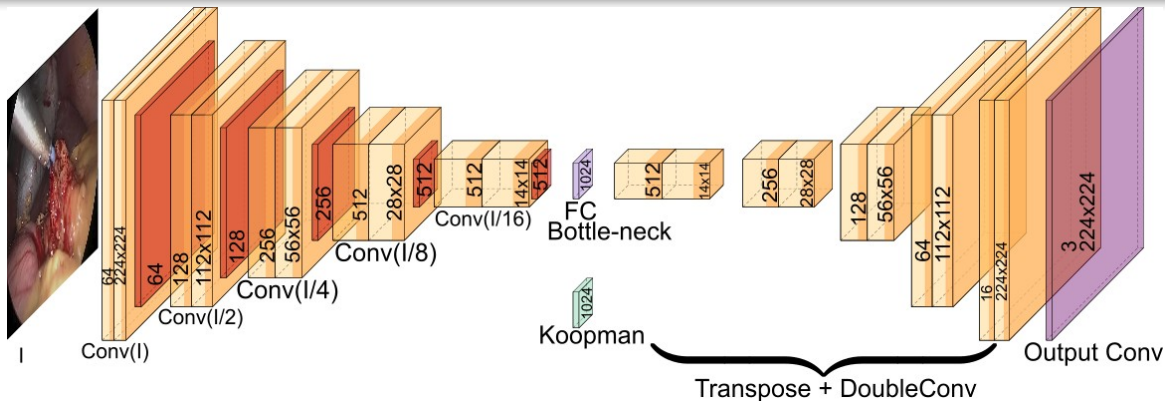




Learning Latent Linear Dynamics from Surgical Videos via Koopman-based Visual Representations

Luke Whatley, Andrei Cozma, Jie Fan, Hairong Qi, Fei Liu (University of Tennessee)



Deep Koopman Operations:

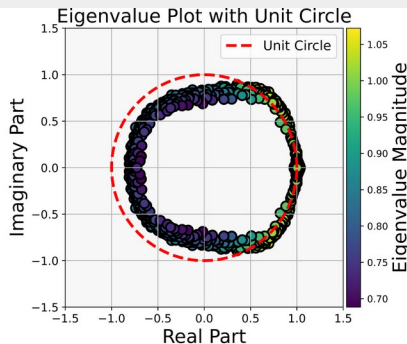
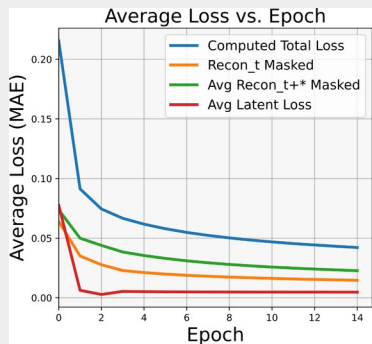
Encoder: $\mathbf{y}_t = \varphi(\mathbf{x}_t)$

Decoder: $\hat{\mathbf{x}}_t = \varphi^{-1}(\mathbf{y}_t)$

Linear lift fun: $\mathcal{K}g(\mathbf{x}) = g(F(\mathbf{x}))$

Prediction: $\hat{\mathbf{x}}_{t+k} = \varphi^{-1}(\mathcal{K}^k \varphi(\mathbf{x}_t))$

Experimental Results



Losses:

Reconstruction:

$$\mathcal{L}_{\text{recon}} = \sum_{t=1}^T \|\varphi^{-1}(\varphi(\mathbf{x}_t)) - \mathbf{x}_t\|_2^2$$

Prediction:

$$\mathcal{L}_{\text{pred}} = \sum_{t=1}^{T-1} \|\varphi^{-1}(\mathcal{K}\varphi(\mathbf{x}_t)) - \mathbf{x}_{t+1}\|_2^2$$

Linearity:

$$\mathcal{L}_{\text{lin}} = \sum_{t=1}^{T-1} \|\varphi(\mathbf{x}_{t+1}) - \mathcal{K}\varphi(\mathbf{x}_t)\|_2^2$$

